# Exercise Sheet 10

## Cascades and Temporal Networks

5942UE Network Science

26 March 2026

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# 10.A Cascades and Threshold Rules

# Exercise 10.A.1: Threshold Cascade by Hand

Graph (edges $A$–$B, B$–$C, A$–$D, C$–$E, D$–$F$) with thresholds: $A(0.4), B(0.5), C(0.3), D(0.5), E(0.5), F(0.3)$
.

Seed at $t = 0$: $A, C$. Synchronous update.

1. Compute the degree of each non-seed node.
2. Trace which nodes adopt at $t = 1, 2, 3$.
3. Does the cascade reach all nodes?

# Exercise 10.A.2: Threshold Heterogeneity

Same graph as 10.A.1. Two experiments:

- **A:** All thresholds raised to $0.6$.
- **B:** Only $B$'s threshold raised to $0.6$ (others as in 10.A.1).

For seed $A, C$:

1. Does the cascade reach all nodes in (A)?
2. Does the cascade reach all nodes in (B)?
3. Why does a single threshold change "block" the cascade in (A)?

# Exercise 10.A.3: Cascade Simulation

Simulate threshold cascades on `nx.barabasi_albert_graph(100, 3)`.

1. At what seed size does the cascade reliably sweep most of the network?
2. Raise threshold to 0.5 — how does final adoption change?
3. Compare seeding **hubs** (top-$k$ degree) vs **random**.

# Exercise 10.A.4: Bridges Differ for Simple vs. Complex Contagion

Dumbbell: two 5-cliques $1..5$ and $6..10$ joined by bridge $5$–$6$. Node 1 adopts first.

1. **Simple contagion** ($\beta = 1$): does it spread to clique B?
2. **Complex contagion** ($\theta = 0.4$): does it cross the bridge?
3. What change would let the cascade cross?

# 10.B Temporal Networks

# Exercise 10.B.1: Order-Dependent Reachability

Three nodes $A, B, C$ with two timed edges:

- $A$–$B$ exists at $t = 1$ only
- $B$–$C$ exists at $t = 2$ only

1. Can information travel $A \to C$? Trace the path.
2. Can it travel $C \to A$?
3. Reverse the timing ($A$–$B$ at $t = 2$, $B$–$C$ at $t = 1$). Does $A \to C$ still work?
4. What does this say about temporal vs. static reachability?

# Exercise 10.B.2: Static Aggregation Hides Causal Order

A hospital contact network over 3 days:

| Day | Edge |
|---|---|
| 1 | Nurse–Patient A |
| 2 | Nurse–Patient B |
| 3 | Doctor–Nurse |

1. Can infection spread from Patient A to Doctor? Trace it.
2. Can it spread from Doctor to Patient A?
3. What might static contact tracing wrongly conclude?

# Exercise 10.B.3: Temporal Reachability in Code

Implement BFS over a temporal edge list.

1. From node 0, which nodes are reachable and at what earliest time?
2. From node 3, is node 0 reachable?
3. Add edge $(3, 0, 5)$. How does reachability change?

# Exercise 10.B.4: When Timing Kills Spreading

Star with hub $H$ and leaves $A, B, C, D$. Disease starts at $A$ at $t = 0$.

- **Scenario X:** all edges active at $t = 1$
- **Scenario Y:** $H$–$A$ at $t = 1$, $H$–$B$ at $t = 2$, $H$–$C$ at $t = 3$, $H$–$D$ at $t = 4$

1. Which nodes get infected in (X)?
2. In (Y)?
3. Now suppose the disease has a **1-step infectious window**. In (Y), does it spread beyond $H$?
4. What does this illustrate about epidemic containment?

# Wrap-Up

- threshold cascades depend on dense local reinforcement, not just connectivity
- bridges are good for simple contagion but bad for complex cascades
- temporal networks make reachability asymmetric and order-dependent
- contact timing relative to infectious windows can decisively contain spread